AI Fact-Checking System

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Abstract

We demonstrate an efficient resource-optimized pipeline involving fact-checking for the climate-based claims executing in the free-tier of the Google Colab kernel runtime avoids any additional services, checkpoints of saved models, or already computed indexes. The system represents the issues of verdict predictions and retrieval of evidence below a constrained budget of compute via four staged parts: One multi-gate sparse retriever which applies a appropriately weighted reciprocal rank fusion over the views of BM25; the multi-scaling feature fusion reranker integrates similarity of sentence embedding and the shallow factual alignment cues involving numeric agreement, lexical overlapping, and the compatibility with the negation; the bounded cross-encoder selector only used on a candidate subset which is prefiltered; and TF-IDF which is a structured-organized logistic classifier which encodes the factual cue tokens and the rank of evidence as input signals that are explicit. In this respective development set, our best Colab-supporting configuration attains a solid evidence retrieval score of 0.2021, claim classification accuracy of 0.5000, harmonic mean of 0.2879, and the classifier macro-F1 score of 0.4659. The point to be noted is that we must safeguard the shallow factual cues and the evidence rank on the way via the pipeline, instead of throwing them through the concatenation of plain evidence, which uniformly enhances the performance of the classifier.

1 Introduction

Climate-related claims usually appear as short statements in news articles, social media posts, or public discussions, while the information needed to verify them is often scattered across a much larger evidence corpus. This makes climate fact-checking different from ordinary claim classification: the model is not only asked to assign a verdict, but must do so based on evidence that has to be found before it can be used. At the same time, a practical system cannot assume unlimited computation, especially in lightweight settings where the full pipeline is expected to run in a free notebook environment such as Google Colab. We therefore frame the project as a resource-constrained evidence-based verification problem (Thorne et al., 2018; Diggelmann et al., 2020). The goal is not to build the strongest possible model in isolation, but to design a pipeline that gives a good balance between evidence recall, verdict accuracy, and runtime cost.

Prior work motivates each stage of this design. FEVER formulates fact checking as joint evidence retrieval and verdict prediction, and CLIMATE-FEVER shows that climate claims add domain-specific retrieval challenges such as contested framing, numerical comparisons, and temporal qualifications (Thorne et al., 2018; Diggelmann et al., 2020). Sparse retrieval methods such as BM25 provide scalable high-recall candidate generation, while reciprocal rank fusion combines heterogeneous lexical views without score calibration (Robertson et al., 1994; Cormack et al., 2009). Dense bi-encoders and sentence embeddings improve semantic matching, and cross-encoders provide stronger claim–evidence interaction for smaller candidate pools (Karpukhin et al., 2020; Reimers and Gurevych, 2019; Nogueira and Cho, 2019). Evidence aggregation and hybrid fact-checking work further suggests that passage structure, lexical alignment, and shallow factual cues carry useful verification signals beyond raw text concatenation (Zhou et al., 2019; Lee et al., 2018; Padia et al., 2018).

Motivated by the resource-constrained setting, our system adopts a staged design in which each module improves evidence quality under a bounded computational budget. First, Multi-Gate Sparse Candidate Retrieval is used to build a high-recall evidence pool: the Word gate captures direct lexical overlap between claims and evidence, the Character gate improves robustness to surface-form and morphological variation, the Structured gate emphasizes claim-critical cues such as entities, numbers, and factual expressions, and the Shared BM25 interaction provides a common sparse scoring basis across these retrieval views (Robertson et al., 1994); Rank fusion then combines the gate outputs into a broader and more stable candidate set than any single gate alone (Cormack et al., 2009). Second, Multi-Scale Feature Fusion reranks the retrieved candidates by combining semantic similarity with shallow factual signals, so the system can benefit from MiniLM-style meaning matching while still preserving precise cues such as lexical overlap, numerical consistency, negation, and retrieval rank (Reimers and Gurevych, 2019). Third, Bounded Neural Evidence Selection applies stronger neural scoring only to a prefiltered subset of candidates, allowing the model to capture richer claim–evidence interactions without applying expensive neural inference to the full corpus (Nogueira and Cho, 2019). Finally, Structured Evidence Classification converts the selected evidence into a compact rank-aware and cue-aware representation for verdict prediction, so the classifier uses both evidence content and retrieval confidence while keeping the full pipeline runnable in the target low-resource environment.

The main contribution of this work is a resource-aware evidence-based verification pipeline that connects retrieval, reranking, evidence selection, and verdict classification under a single executable framework. Rather than optimizing each component in isolation, the system is designed around the performance–runtime trade-off required by low-resource execution. Within this pipeline, the key technical contributions are Multi-Gate Sparse Candidate Retrieval, which improves candidate recall by combining complementary sparse views; Multi-Scale Feature Fusion, which integrates semantic similarity with shallow factual signals for more reliable reranking; Bounded Neural Evidence Selection, which limits expensive neural scoring to the most promising candidates; and Structured Evidence Classification, which represents selected evidence in a form that preserves rank and factual cues for final prediction. In addition, we contribute a set of diagnostic experiments that analyze the effect of sparse gates, rank fusion, neural selection, evidence representation, and classifier sensitivity, providing insight into which stages most strongly affect evidence quality, label accuracy, and runtime.

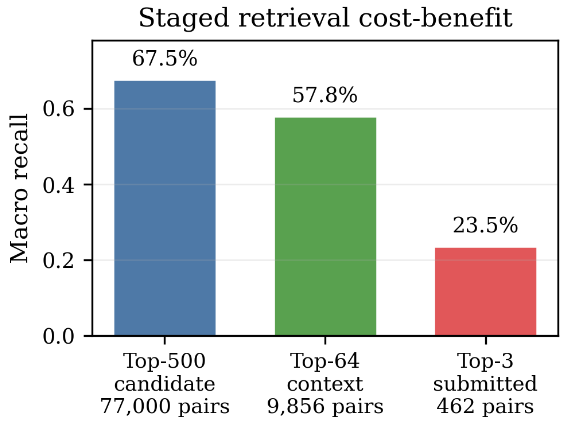

Figure 1: Staged pruning keeps recall while reducing pair count. The candidate and context pools retain most recoverable evidence by using neural scoring on progressively smaller pools; full-corpus neural scoring would require about 186 million development pairs.

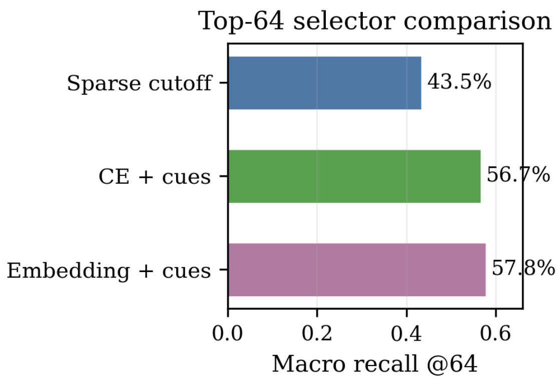

Figure 2: Embedding plus factual cues is the best top-64 selector, reaching 57.8% macro recall above sparse cutoff and cross-encoder-plus-cue variants.

2 Method

2.1 Pipeline Overview

Let $\mathcal{C}$ be claims and $\mathcal{E}$ evidence. The output is a verdict and a small evidence set per claim. Since scoring all $(c, e) \in \mathcal{C} \times \mathcal{E}$ pairs is infeasible, we use a staged pipeline:

$$\mathcal{E} \to \mathcal{A}(c) \to \mathcal{B}(c) \to \mathcal{D}(c),$$

where $\mathcal{A}(c)$ is a broad sparse pool, $\mathcal{B}(c)$ a reranked context pool, and $\mathcal{D}(c)$ the submitted evidence. Cheap stages maximize recall; semantic models operate on progressively smaller pools. The four components are multi-gate sparse retrieval, multi-scale reranking, bounded cross-encoder selection, and structured-evidence classification.

2.2 Multi-Gate Sparse Candidate Retrieval

The sparse stage maximizes recall by applying one BM25 scoring operator to multiple feature spaces (Robertson et al., 1994). The four gates differ in analyzer $a_g(\cdot)$ and vocabulary V_g. Let $q_g(c)$ and $x_g(d)$ be the resulting count vectors for claim c and evidence document d. For each gate, the notebook builds a claim matrix Q_g and an evidence matrix X_g, reweights the evidence matrix with BM25, and computes batched sparse products $Q_g X_g^\top$ before retaining the top-ranked evidence rows per claim.

Word gate. Word terms give the direct lexical view: content-word overlap raises scores and IDF weighting emphasizes discriminative claim terms. The analyzer removes English stop words and uses both unigrams and adjacent word bigrams, so phrase matches such as scientific terms or named processes can contribute as single features.

Character gate. Character n-grams capture partial overlap, inflection, hyphenation, and spelling variation missed by exact words. This gate uses word-boundary character n-grams, making it useful for abbreviations, compounds, and morphology in climate terminology.

Structured gate. Structured cues keep numbers, years, salient capitalized terms, and negation markers. These are encoded as typed tokens such as year, number, entity-like, and negation features, so numeric or negation agreement can affect retrieval even when surrounding words differ.

Pseudo-relevance feedback gate. PRF expands the word query using high-ranked feedback evidence. If $\mathcal{F}(c)$ is the feedback set,

$$q_{\mathrm{prf}}(c) = q_{\mathrm{word}}(c) + \eta \sum_{d \in \mathcal{F}(c)} w(c,d) x_{\mathrm{word}}(d),$$

where $w(c,d)$ downweights lower-ranked feedback documents. PRF adds related terms absent from the original claim while staying in the sparse word space.

Shared BM25 interaction. After feature construction, each gate applies the same BM25 interaction. For term t in document d,

$$\mathrm{idf}_t = \log\left(\frac{N - df_t + 0.5}{df_t + 0.5} + 1\right),$$

$$\mathrm{BM25}_{d,t} = \mathrm{idf}_t \frac{(k_1 + 1) f_{d,t}}{f_{d,t} + k_1(1 - b + b|d|/\overline{|d|})}.$$

The inverse-document-frequency factor suppresses common features, so rare claim-specific evidence terms contribute more than corpus-wide background terms. The query side remains a count vector, giving the sparse dot product

$$s_g(c,d) = q_g(c)^\top \mathrm{BM25}_g(d).$$

Thus g chooses the representation, and BM25 weights it. The notebook uses SciPy/NumPy sparse matrices, so document-frequency counts, BM25 reweighting, and query–document dot products run in optimized C/C++ kernels and vectorized array operations.

Rank fusion. Each gate produces a ranked list. The fifth sparse view is their weighted RRF fusion (Cormack et al., 2009):

$$S_{\mathrm{RRF}}(c,d) = \sum_{g \in G} \frac{\lambda_g}{K_{\mathrm{RRF}} + r_g(c,d)},$$

where missing documents contribute zero. We treat λ_g as a reliability prior: stronger standalone gates receive larger weights, and complementary factual gates retain smaller weights. The structured gate contributes complementary recall in fusion, as shown by ablation; RRF also removes the need for cross-gate score calibration.

Figures 1 and 3 justify the candidate stage: fusion improves top-500 recall over any single source, and staging concentrates neural scoring on high-value candidate pools.

2.3 Multi-Scale Feature Fusion

Reranking fuses dense similarity, sparse source rank, word overlap, numeric agreement, negation compatibility, and length shape. MiniLM embeddings are mean-pooled and ℓ_2-normalized, following sentence-embedding retrieval practice (Reimers and Gurevych, 2019), so cosine similarity is an inner product:

$$s_{\mathrm{emb}}(c,d) = h(c)^\top h(d).$$

The shallow factual features are summarized as

$$\phi(c,d) = [\phi_{\mathrm{rank}}, \phi_{\mathrm{lex}}, \phi_{\mathrm{num}}, \phi_{\mathrm{neg}}, \phi_{\mathrm{len}}, s_{\mathrm{emb}}].$$

Here ϕ_{rank} is the reciprocal sparse-source rank, ϕ_{lex} contains claim-normalized and evidence-normalized word overlap, ϕ_{num} counts shared numeric strings, ϕ_{neg} marks negation compatibility, and ϕ_{len} records normalized character-length difference. A lightweight gradient-boosted matcher is

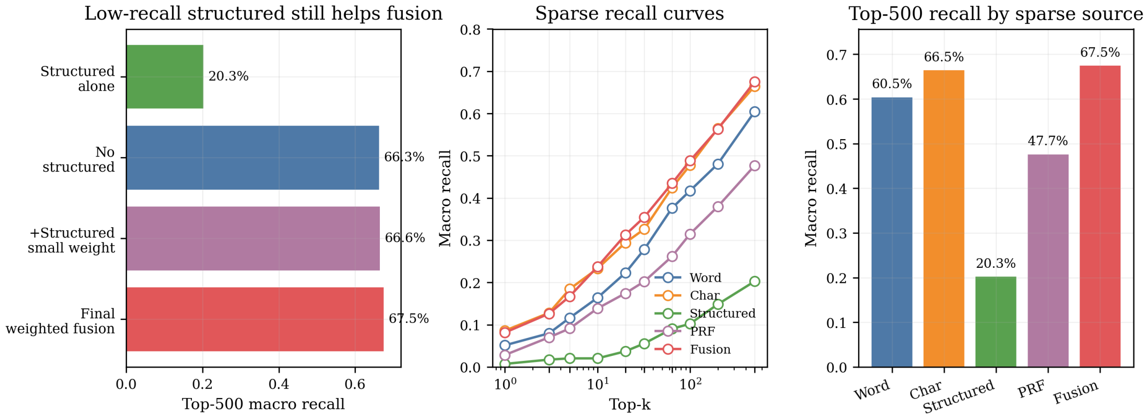

Figure 3: Sparse fusion gives the best top-500 recall and retains structured cues for complementarity. Character matching is the strongest single source; structured cues improve fused recall under controlled ablation, so the final system keeps a small structured weight.

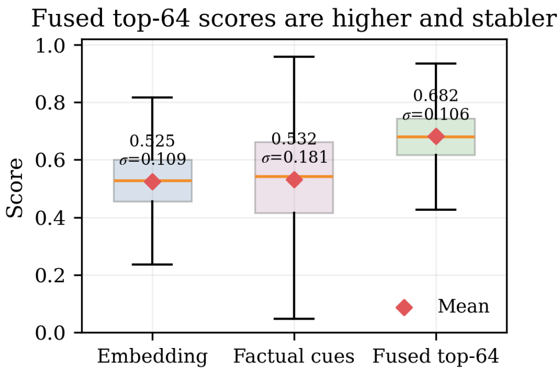

Figure 4: Top-64 fusion raises the retained-candidate score distribution and reduces dispersion. In the Colab diagnostic, fused scores have mean/std = 0.682/0.106, compared with 0.525/0.109 for embeddings and 0.532/0.181 for factual cues.

trained on gold claim–evidence pairs and sampled non-gold candidates from the training claims. It estimates $p_{\text{feat}}(c, d)$, the probability that a candidate is useful evidence under these shallow signals. The context score is

$$S_{\text{ctx}}(c, d) = \alpha\, \widetilde{s}_{\text{emb}}(c, d) + (1 - \alpha)\, p_{\text{feat}}(c, d),$$

where $\widetilde{s}_{\text{emb}}$ is claim-wise normalized. Embeddings handle semantic relatedness; factual features guard against number, negation, and lexical-grounding errors.

2.4 Bounded Neural Evidence Selection

The submitted evidence stage uses a stricter operating point than the classifier context. Cross-encoders jointly encode claim and evidence (Nogueira and Cho, 2019), so we reserve them for the sparse/embedding-prefiltered pool. Candidate evidence text is truncated before tokenization, then claim–evidence pairs are scored in PyTorch batches. For each claim, raw cross-encoder logits and embedding similarities are min–max normalized over the candidate pool. The final evidence score fuses cross-encoder confidence, embedding confidence, embedding rank, and sparse source rank:

$$\begin{aligned} S_{\text{sub}}(c, d) &= \beta_1 \widetilde{s}_{\text{CE}} + \beta_2 \widetilde{s}_{\text{emb}} \\ &\quad + \frac{\beta_3}{\tau + r_{\text{emb}}} + \frac{\beta_4}{\tau + r_{\text{src}}}. \end{aligned}$$

Coefficients are fixed. Candidates outside the cross-encoder subset remain ranked by embedding and source-rank terms, which keeps neural cost bounded and retains useful sparse candidates.

Figures 2, 4, and 5 justify multi-scale fusion: factual cues act as complementary signals that improve semantic ranking and stabilize the retained candidate pool.

2.5 Structured Evidence Classification

The classifier predicts from a bounded ordered evidence context:

$$x_c = [\text{claim}; z_1, z_1, \ldots, z_m, z_m, z_{m+1}, \ldots, z_L],$$

where the highest-ranked evidence lines are repeated to raise their TF-IDF weight. The leading evidence receives a larger text budget, while later rows are truncated so the context keeps more distinct evidence items. Each evidence line also gets a rank prefix plus cue tokens: a word-overlap bucket, number match or mismatch status, claim-number-only status, and negation compatibility. The logistic model learns weights for these cue tokens from training data. Let v_c be the TF-IDF vector; a one-vs-rest logistic model learns one binary classifier

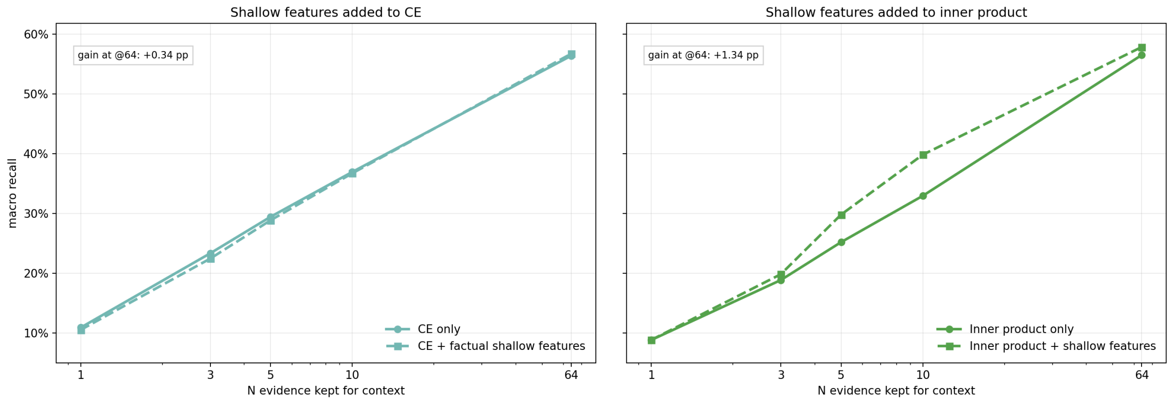

Figure 5: Factual cues preserve or improve semantic ranking. Cue gains are larger for embedding inner product and smaller for the already strong cross-encoder, supporting feature fusion over a purely semantic selector.

per label:

$$P(y = k \mid c) = \sigma(\theta_k^\top v_c + b_k).$$

The largest decision score gives the label. This keeps classification cheap while preserving retrieval rank and factual signals.

3 Results

3.1 Main Experimental Results

The final system achieved an evidence retrieval F-score of 0.2021, claim classification accuracy of 0.5000, a harmonic mean score of 0.2879, and a classifier macro-F1 score of 0.4659, as shown in Table 2. Overall, these results show that the system was able to balance retrieval performance and classification accuracy while still remaining lightweight enough to run within free Google Colab limitations. The sparse retrieval experiments showed that combining multiple sparse retrieval methods improved evidence recall compared to using only a single retrieval source. Figure 3 shows that the weighted reciprocal rank fusion retriever achieved the highest top-500 macro recall of 67.5%, outperforming all individual sparse retrieval gates. Character-based retrieval achieved the strongest standalone recall, while the structured retrieval gate still improved the final fusion results by adding complementary factual information.

The staged retrieval pipeline also reduced computational cost while still preserving useful evidence candidates. As shown in Table 1 and Figure 6, recall gradually decreased as the candidate pool became smaller across each retrieval stage. The sparse top-500 stage achieved 67.5% macro recall, while the reranked top-64 context pool retained 57.8% macro recall. After the final top-3 evidence selection stage, the submitted evidence set achieved an evidence retrieval F-score of 0.2021. The reranking experiments further showed that combining semantic embeddings with shallow factual cues improved evidence selection quality. Figure 2 shows that the embedding-plus-cues selector achieved the highest top-64 macro recall of 57.8%, outperforming both the sparse cutoff baseline and the cross-encoder-plus-cues configuration. Figure 4 also shows that the fused retrieval scores produced a more stable retained candidate distribution compared to using embeddings or factual cues alone. The classifier diagnostics in Figure 7 show that the model performed best on the SUPPORTS class, while most classification errors occurred between SUPPORTS, REFUTES and NOT-ENOUGH-INFO. The DISPUTED class was the hardest class to classify due to the smaller number of development examples and more ambiguous evidence relationships.

3.2 Ablation Study

The ablation experiments showed that both factual feature fusion and structured evidence formatting improved downstream classification performance. Figure 5 shows that adding shallow factual cues consistently improved semantic ranking performance for both embedding similarity and cross-encoder retrieval. The improvements were more noticeable for embedding similarity, suggesting that lexical overlap, number agreement and negation matching helped preserve stronger factual consistency between claims and evidence. The classifier ablation experiments in Table 3 also showed that structured evidence formatting performed better than plain evidence concatenation. The plain context representation achieved an accuracy of 0.4351

Stage	F-score	Macro recall	Hit-any
Sparse top-500	0.0083	0.6754	0.9026
Context top-64	0.0517	0.5784	0.8636
Submitted top-3	0.2021	0.2355	0.4675

Table 1: Retrieval metrics from the self-generated notebook artifacts.

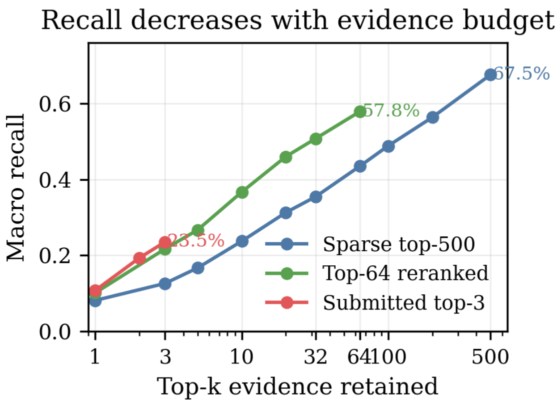

Figure 6: Recall-vs-K curves show the pruning trade-off behind the selected operating points. The sparse pool keeps broad top-500 recall, the reranked pool preserves most recoverable evidence at top-64, and the submitted top-3 is the strict evidence set used for scoring.

and macro-F1 score of 0.3762, while the enhanced context representation improved performance to 0.4935 accuracy and 0.4544 macro-F1 at C = 0.25. The best overall development performance was achieved using the enhanced representation with C = 0.125, reaching 0.5195 accuracy and 0.4841 macro-F1.

3.3 Sensitivity Analysis

The retrieval sensitivity analysis showed that evidence recall was highly dependent on the retained evidence budget. Figure 6 shows that retrieval recall gradually decreased as fewer evidence candidates were retained. However, the staged retrieval architecture was still able to preserve most recoverable evidence while significantly reducing the number of claim-evidence pairs requiring neural scoring. The classifier sensitivity analysis in Table 3 showed that logistic regression regularisation strength affected classification performance. Lower C values achieved stronger generalisation performance, with the best macro-F1 score achieved at C = 0.125. Performance gradually decreased as C increased, suggesting that stronger regularisation helped reduce overfitting within the structured evidence representation.

Metric	Value
Evidence Retrieval F-score (F)	0.2021
Claim Classification Accuracy (A)	0.5000
Harmonic Mean of F and A (H)	0.2879
Top-3 Macro Recall	0.2355
Classifier Macro-F1	0.4659

Table 2: Final Colab-reproduced development result. F, A, and $H = 2FA/(F + A)$ are official course metrics; the last two rows summarize retrieval and classification behavior.

4 Discussion and Critical Analysis

The results showed that evidence retrieval quality had a major impact on overall performance. Better retrieval generally led to stronger classification because the classifier relied on receiving relevant evidence. The staged retrieval architecture therefore worked well for the resource-constrained setting: it preserved useful candidates while reducing the pool before applying more expensive neural reranking.

Sparse retrieval remained important even when semantic reranking was used later. Character-based retrieval was especially strong because climate claims often contain scientific terminology, abbreviations, and numeric expressions that benefit from partial lexical matching. Reciprocal rank fusion improved recall by combining complementary sparse evidence relationships. The reranking experiments also showed that semantic similarity alone was not enough: shallow factual cues such as lexical overlap, number agreement, and negation matching helped preserve factual consistency between claims and evidence.

The classifier ablations showed the same pattern downstream. Enhanced evidence formatting preserved rank information and factual cue tokens, allowing the TF-IDF logistic regression classifier to learn stronger patterns than plain concatenation. The main limitation is still evidence selection: exact gold evidence retrieval remains difficult, and the DISPUTED class is hard because it has fewer examples and requires reasoning over more ambiguous evidence relationships.

5 Conclusion

This paper examined how to balance claim-label accuracy, evidence recall, and runtime for climate claim fact-checking under free Google Colab constraints. The final pipeline combines multi-gate sparse retrieval, feature-fusion reranking, bounded cross-encoder evidence selection, and structured TF-IDF classification. On the development set, it achieved evidence F-score 0.2021, claim accuracy 0.5000, harmonic mean 0.2879, and classifier macro-F1 0.4659.

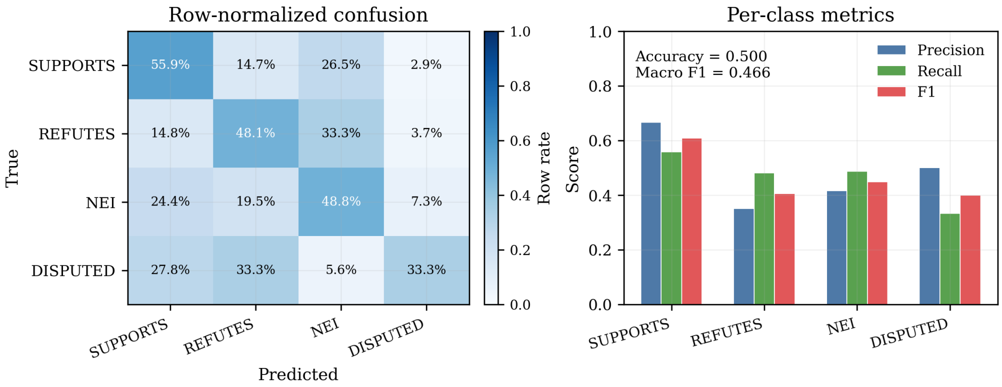

Figure 7: Classifier diagnostics. Row-normalized confusion controls for class support, and per-class precision/recall/F1 summarizes the classification report.

Evidence input variant	C	Acc.	Macro-F1
Plain evidence text	0.25	0.4351	0.3762
Add evidence rank token	0.25	0.4351	0.3749
Repeat top evidence	0.25	0.4416	0.3938
Rank-weighted text budget	0.25	0.4351	0.3882
Add factual cue tokens	0.25	0.4935	0.4544
Factual cues + tuned C	0.125	0.5195	0.4841

LogReg C	Acc.	Macro-F1
0.125	0.5195	0.4841
0.25	0.4935	0.4544
0.5	0.4740	0.4263
1.0	0.4675	0.4184
2.0	0.4416	0.3825
4.0	0.4545	0.3913
8.0	0.4481	0.3925

Table 3: Classifier ablations. The left block compares how retrieved evidence is formatted before TF-IDF classification. The right block keeps the factual cue format fixed and sweeps logistic-regression regularization.

The main conclusion is that retrieval signals should be carried forward rather than discarded. Multi-gate sparse fusion reached 67.5% macro recall at top-500, shallow factual cues improved semantic reranking, and structured evidence input improved classification over plain concatenation. The remaining bottleneck is top-3 selection, where many exact gold passages are still lost before final submission, especially for DISPUTED claims.

Team contributions. Kaitlyn worked on results, discussion, introduction, and code. Jun worked on imagery, introduction, references, method, and code. Om worked on the abstract, related work, conclusion, references, and code.

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